

Toward Drought-Sensitive Methane Fluxes: Evidence from Boreal Peatlands

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Abstract

We examine the behaviour of estuarine salinity in coastal wetlands. Residual error grows approximately linearly with the number of coupled parameters. The reduced model retains most of the explanatory power at a fraction of the computational cost. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

1. Introduction

Calibration drift over the trial period remained below the threshold set out in the protocol. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Neither of the alternative formulations improved on the simple additive model in our setting. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

Taken together, these findings argue for treating the two regimes separately in future analyses. Neither of the alternative formulations improved on the simple additive model in our setting. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Residual error grows approximately linearly with the number of coupled parameters. We release the processing scripts and intermediate artefacts to support independent verification. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

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2. Related Work

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3. Materials and Methods

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separately in future analyses. Longer observation windows narrow the confidence interval but do not shift the central estimate. Replicate agreement was high, suggesting the protocol is robust to small handling differences.

4. Experimental Setup

The measurements reported here were collected over twelve independent trials under controlled conditions. Longer observation windows narrow the confidence interval but do not shift the central estimate. Neither of the alternative formulations improved on the simple additive model in our setting. The reduced model retains most of the explanatory power at a fraction of the computational cost. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Calibration drift over the trial period remained below the threshold set out in the protocol. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Sensitivity to the smoothing window proved modest across the range we examined. Replicate agreement was high, suggesting the protocol is robust to small handling differences.

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5. Results

Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. We release the processing scripts and intermediate artefacts to support independent verification. Residual error grows approximately linearly with the number of coupled parameters. The measurements reported here were collected over twelve independent trials under controlled conditions. Sensitivity to the smoothing window proved modest across the range we examined. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

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6. Discussion

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7. Limitations

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8. Conclusion

Longer observation windows narrow the confidence interval but do not shift the central estimate. Neither of the alternative formulations improved on the simple additive model in our setting. We release the processing scripts and intermediate artefacts to support independent verification. Calibration drift over the trial period remained below the threshold set out in the protocol. The measurements reported here were collected over twelve independent trials under controlled conditions. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Taken together, these findings argue for treating the two regimes separately in future analyses. Sensitivity to the smoothing window proved modest across the range we examined.

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